

LESSON 25 / 第二十五課

Urban Trees and Heat: From City-Scale Cooling to Leaf Energy Balance

城市樹木與熱：從城市尺度冷卻到葉片能量平衡

Series note. Lessons 23–24 treated urban mortality rates and drought-induced death as physiological endpoints. This lesson treats the complementary service those same trees are asked to provide: cooling a city that is itself a heat engine. The organising principle remains scale.

系列說明。第二十三至二十四課把城市死亡率與乾旱致死當成生理終末。本課處理同一批樹木被要求提供的互補服務：冷卻一座本身就是熱機的城市。組織原則仍是尺度。

Abstract 摘要

Urban trees cool cities by two physically distinct routes — shade, which intercepts shortwave radiation before it reaches pavement and walls, and transpiration, which converts absorbed net radiation into latent heat. A third, architectural route is the modification of near-surface wind. These routes do not substitute for one another, and they do not scale linearly with canopy cover. Across 8,919 large urban areas, McDonald et al. [1] estimated that existing tree cover mitigates 41–49% of the maximum potential air-temperature urban heat island that would occur with no canopy, yet the population-weighted mean summer cooling is only 0.15 ± 0.03 °C, and current or plausible future canopy offsets only about 10% of mid-century climate-change warming under a moderate emissions pathway. Guo et al. [2], working with land-surface temperature across 761 megacities, showed that the sign of the vegetation effect reverses with aridity: in 22% of cities with annual precipitation below 1,000 mm, grassland and cropland are net warming surfaces, and during extreme heat trees fail to cool 25% of cities. At block scale, Ziter et al. [3] measured a nonlinear air-temperature response, with the steepest daytime cooling once canopy exceeded about 40% within a 60–90 m radius. Rahman et al. [4] identified canopy density as the strongest trait predictor of both surface-temperature reduction and transpirational cooling, and reported that trees over grass transpired an order of magnitude more than trees in paved cut-out pits. Pataki et al. [5] showed that the water bill of that latent-heat pathway is species-specific over nearly two orders of magnitude in a single irrigated metropolis. At the leaf, Leuzinger and colleagues [6,7] found that crown temperature is not air temperature: large-leaved, dense-crowned street trees ran several kelvin hotter than park trees or fine-leaved species under the same ambient air. The review therefore treats cooling as a contingent energy-balance outcome, not a guaranteed ecosystem service, and as a process that shares its water and stomatal account with the drought-mortality mechanisms of Lesson 24.

城市樹木以兩條物理上不同的路徑冷卻城市——遮蔭，在短波輻射到達鋪面與牆體之前攔截它；以及蒸騰，把吸收的淨輻射轉為潛熱。第三條建築性路徑是近地表風場的改變。這三條路徑不能互相替代，也不隨樹冠覆蓋線性放大。McDonald 等 [1] 在 8,919 個大型城市區估計，現有樹冠可緩解「無樹冠時最大可能氣溫熱島」的 41–49%，但人口加權的夏季平均冷卻僅 0.15 ± 0.03 °C，而現有或可預期的未來樹冠，在中度排放路徑下只能抵消本世紀中葉氣候變暖的約 10%。Guo 等 [2] 以 761 個特大城市的地表溫度顯示，植被效應的符號隨乾旱度反轉：年降水低於 1,000 mm 的城市中有 22%，草地與農地是淨增溫表面；極端高溫期間，樹木在 25% 的城市未能冷卻。街廓尺度上，Ziter 等 [3] 測得非線性氣溫響應，當 60–90 m 半徑內樹冠超過約 40% 時日間冷卻最陡。Rahman 等 [4] 指出樹冠密度是地表降溫與蒸騰冷卻最強的性狀預測因子，且種在草地上的樹蒸騰量比鋪面樹穴中的樹高出一個數量級。Pataki 等 [5] 顯示，該潛熱路徑的水賬在同一灌溉都市內可因樹種相差近兩個數量級。在葉片尺度，Leuzinger 等 [6,7] 發現樹冠溫度不是氣溫：大葉、密冠的行道樹在相同環境氣溫下，可比公園樹或細葉樹種高出數個開爾文。本綜述因此把冷卻當成有條件的能量平衡結果，而不是保證交付的生態系統服務，並且與第二十四課的乾旱死亡機制共用水分與氣孔賬戶。

1. Scope of the review 1. 綜述範圍

The object is the thermal effect of urban trees, not a planting programme and not a single-city case file. Heat is treated as an energy-balance problem that is measured differently at each scale (Fig. 1). Land-surface temperature (LST) from satellites is not air temperature at 1.5–2 m, and neither is mean radiant temperature as felt by a

pedestrian. Alonzo, Ibsen and Locke [8] mapped 115 studies (2018–2024) and recorded that conflicting conclusions often follow from this metric mismatch. Section 2 treats global and city-scale air- and surface-temperature results. Section 3 treats the block-scale canopy threshold. Section 4 separates the three cooling pathways. Section 5 treats the water cost of latent-heat cooling. Section 6 treats the leaf. Section 7 records when the service fails. Study questions test whether the numbers, not the slogan that “trees cool cities,” have been retained.

對象是城市樹木的熱效應，不是種植計畫，也不是單一城市檔案。熱被當成能量平衡問題，在每一尺度以不同量測定義（圖1）。衛星地表溫度（LST）不是 1.5–2 m 氣溫，後者也不是行人感受到的平均輻射溫度。Alonzo、Ibsen 與 Locke [8] 整理 2018–2024 年 115 項研究，記錄許多互相衝突的結論來自這一量測錯位。第 2 節處理全球與城市尺度的氣溫與地表溫度結果。第 3 節處理街廓尺度樹冠閾值。第 4 節分開三條冷卻路徑。第 5 節處理潛熱冷卻的水成本。第 6 節處理葉片。第 7 節記錄服務何時失效。文末習題檢驗保留的是數字，而不是「樹木冷卻城市」的口號。

Figure 1. Nested scales used in this review 圖 1. 本綜述所用嵌套尺度



Figure 1. Nested scales used to organise this review, from city-scale UHI to the leaf energy balance. 圖 1. 本綜述的嵌套尺度：由城市熱島到葉片能量平衡。

2. City and global patterns 2. 城市與全球格局

The urban heat island (UHI) is the difference between urban and rural temperature under a specified weather envelope. It is not a property of trees. Trees enter the account as one of several modifiers of the urban energy balance: they intercept shortwave, they evaporate water, they change roughness, and they occupy volume that would otherwise be paved or built [9]. McDonald et al. [1] asked a counterfactual that the earlier literature rarely quantified at global scale: how large would the air-temperature UHI be if the same 8,919 functional urban areas had no tree canopy? Current canopy was estimated to remove 41–49% of that maximum potential UHI. The population-weighted mean summer air-temperature reduction was 0.15 ± 0.03 °C, with a range of 0.0–2.7 °C across cities. About 914 million people (95% CI 805–1,040 million) received cooling greater than 0.25 °C. The same paper found that cooling is larger in already cooler places — high-income countries and suburbs — and that current or plausible future tree cover would offset only about 10% (6.7–18% current; 6.3–17% future) of median mid-century warming under a moderate emissions scenario. Trees therefore halve a small urban increment; they do not cancel the climate trend.

城市熱島（UHI）是指定天氣包絡下城、鄉溫度之差。它不是樹木的屬性。樹木作為城市能量平衡的若干修飾因子之一進入賬戶：攔截短波、蒸發水分、改變粗糙度，並佔據否則會被鋪面或建物占據的體積 [9]。McDonald 等 [1] 提出早期文獻很少在全球尺度定量的反事實：若同一批 8,919 個功能城市區沒有樹冠，氣溫熱島會有多大？現有樹冠估計可去掉該最大可能熱島的 41–49%。人口加權的夏季氣溫降幅為 0.15 ± 0.03 °C，城市間範圍 0.0–2.7 °C。約 9.14 億人（95% CI 8.05–10.40 億）獲得大於 0.25 °C 的冷卻。同一論文發現，冷卻在本已較涼之處更大——高收入國家與郊區——而現有或可預期的未來樹冠，只能抵消中度排放情境下本世紀中葉暖化中位數的約 10%（現況 6.7–18%；未來 6.3–17%）。樹木因此是把一個偏小的城市增量減半；它們並不取消氣候趨勢。

Guo et al. [2] shifted the metric to LST and the sample to 761 megacities in 105 countries. Across that set, urban trees were cooler than built-up surfaces in 98% of cities, with a mean surface-temperature contrast of -3.71 ± 1.73 °C, larger than grassland (-1.44 ± 1.91 °C) or cropland (-1.86 ± 2.04 °C). The sign was not universal. In 22% of cities with annual precipitation below 1,000 mm, vegetation — particularly grass and crops — produced net warming relative to built surfaces, because lower albedo and reduced heat storage outweighed the limited evapotranspiration. During extreme heat, trees failed to cool 25% of cities; grass and crops failed in 71% and 82%. The mechanism given is reduced evapotranspiration under high vapour-pressure deficit and impeded canopy conductance [2]. That sentence is the same coupling Lesson 24 recorded for mortality: VPD accelerates depletion of the water account that latent-heat cooling requires [10,11].

Guo 等 [2] 把量測改為 LST，樣本改為 105 國 761 個特大城市。在該集合中，城市樹木在 98% 的城市比建成表面更涼，平均地表溫度對比 -3.71 ± 1.73 °C，大於草地 (-1.44 ± 1.91 °C) 或農地 (-1.86 ± 2.04 °C)。符號並非普遍。年降水低於 1,000 mm 的城市中有 22%，植被——尤其是草與作物——相對建成表面淨增溫，因為較低反照率與減少的熱貯存壓過了有限的蒸散。極端高溫期間，樹木在 25% 的城市未能冷卻；草地與農地分別在 71% 與 82% 失效。給出的機制是高水汽壓差下蒸散下降與冠層導度受阻 [2]。這句話與第二十四課記錄的死亡耦合相同：VPD 加速潛熱冷卻所依賴的水分賬戶耗竭 [10,11]。

Alonzo et al. [8] supply the map of the recent literature rather than a new global estimate. Trees outperform other vegetation when water is available, especially in hotter and drier climates; dense, tall canopies cool at broader scales; mixed plantings with shrubs or grass enhance local effects; arid and tropical zones remain under-sampled; and LST versus air temperature continues to generate opposite headlines from the same canopy.

Alonzo 等 [8] 提供的是近期文獻地圖，而不是新的全球估計。有水時樹木優於其他植被，在更熱更乾的氣候尤其如此；密而高的樹冠在較大尺度冷卻；與灌木或草混植增強局地效應；乾旱與熱帶區仍取樣不足；LST 對氣溫繼續讓同一片樹冠產生相反標題。

3. The block-scale threshold 3. 街廓尺度閾值

City-mean canopy is a poor predictor of what a pedestrian experiences. Ziter et al. [3] rode instrumented bicycles through Madison, Wisconsin, sampling air temperature every 5 m along ten transects that spanned 0–100% canopy and 0–100% impervious cover. Within-city daytime air temperature varied by a mean of 3.5 °C (range 1.1–5.7 °C). Temperature fell nonlinearly with canopy. The steepest cooling appeared once canopy exceeded about 40%, and the effect strengthened when land cover was summarised at 60–90 m — a typical city-block radius — rather than at 10 or 30 m. Daytime air temperature rose linearly with impervious cover, but the warming per unit impervious area was smaller than the cooling per unit canopy. Night-time variation was smaller (mean 2.1 °C) and tracked impervious cover more than trees, consistent with storage and longwave emission from mass rather than with daytime shade.

城市平均樹冠很難預測行人所經歷的。Ziter 等 [3] 在威斯康辛州麥迪遜以儀器自行車取樣，沿十條樣線每 5 m 測氣溫，覆蓋 0–100% 樹冠與 0–100% 不透水表面。城內日間氣溫平均變異 3.5 °C（範圍 1.1–5.7 °C）。溫度隨樹冠非線性下降。最陡的冷卻出現在樹冠超過約 40% 之後，且當土地覆蓋以 60–90 m（典型街廓半徑）而非 10 或 30 m 匯總時效應更強。日間氣溫隨不透水覆蓋線性上升，但單位不透水面積的增溫小於單位樹冠的降溫。夜間變異較小（平均 2.1 °C），較追隨不透水覆蓋而非樹木，與質量的貯熱與長波放射一致，而不是與日間遮蔭一致。

Two operational consequences follow. First, a city-wide 20% canopy target can leave every individual block below the threshold at which air-temperature cooling becomes large. Second, the relevant management unit for pedestrian heat is the block, not the municipal mean. Figure 4A is a schematic of that nonlinearity; it is not a universal calibration curve. Madison is a humid-continental city. The 40% figure should be treated as a measured local threshold, not as a planting specification for arid or tropical streets.

由此得出兩項操作後果。第一，全市 20% 樹冠目標可以讓每一個街廓都低於氣溫冷卻變大的閾值。第二，行人熱的相關管理單元是街廓，不是市均值。圖 4A 是該非線性的示意，不是通用率定曲線。麥迪遜是濕潤大陸性城市。40% 應被當成測得的局地閾值，而不是乾旱或熱帶街道的種植規格。

4. Three pathways that do not substitute 4. 不可替代的三條路徑

Rahman et al. [4] meta-analysed empirical, ground-level measurements of below-canopy surface temperature and transpirational (air) cooling. Canopy density was the strongest driver of both. Surface-temperature reduction increased almost linearly with canopy density, but the slope was shallower over transpiring grass than over asphalt. Climate of the host city mattered: oceanic and continental sites showed larger air-temperature cooling than Mediterranean sites, and subtropical sites showed weaker surface-temperature cooling. Among traits, dark, thin leaves (< 0.15 mm) and diffuse-porous wood anatomy associated with stronger cooling. The review also recorded a protocol problem: tropical and subtropical measurements remain sparse, and study designs are not standardised.

Rahman 等 [4] 對冠下地表溫度與蒸騰（氣溫）冷卻的地面實測做了統合分析。樹冠密度是兩者最強驅動因子。地表降溫幾乎隨樹冠密度線性增加，但在蒸騰草地上的斜率比在瀝青上平。宿主城市氣候有關：海洋性與大陸性地點的氣溫冷卻大於地中海地點，亞熱帶地點的地表降溫較弱。性狀方面，深色、薄葉（ < 0.15 mm）與散孔材解剖與更強冷卻相關。該綜述也記錄了規程問題：熱帶與亞熱帶測量仍稀少，研究設計未標準化。

Winbourne et al. [12] separated the two main routes mechanistically. Shade is a morphological property of crown projection, leaf-area density and solar geometry; it operates whether or not the tree is transpiring. Transpiration is a physiological flux controlled by stomatal conductance, soil-water potential, vapour-pressure deficit and the built-environment radiation load. The two can be decoupled: a droughted tree still casts a shadow and can still heat the air if its leaves run hotter than ambient. A third route — modification of the wind field — is a property of the street canyon and of crown architecture, not of leaf physiology [9,12]. Figure 2 is the operational distinction.

Winbourne 等 [12] 從機制上分開兩條主路徑。遮蔭是樹冠投影、葉面積密度與太陽幾何的形態屬性；無論樹木是否在蒸騰都運作。蒸騰是由氣孔導度、土壤水勢、水汽壓差與建成環境輻射負荷控制的生理通量。兩者可以脫鉤：受旱的樹仍投影，若葉溫高於環境氣溫仍可使空氣增熱。第三條路徑——風場修飾——是街道峽谷與樹冠架構的屬性，不是葉生理的屬性 [9,12]。圖 2 即該操作區分。

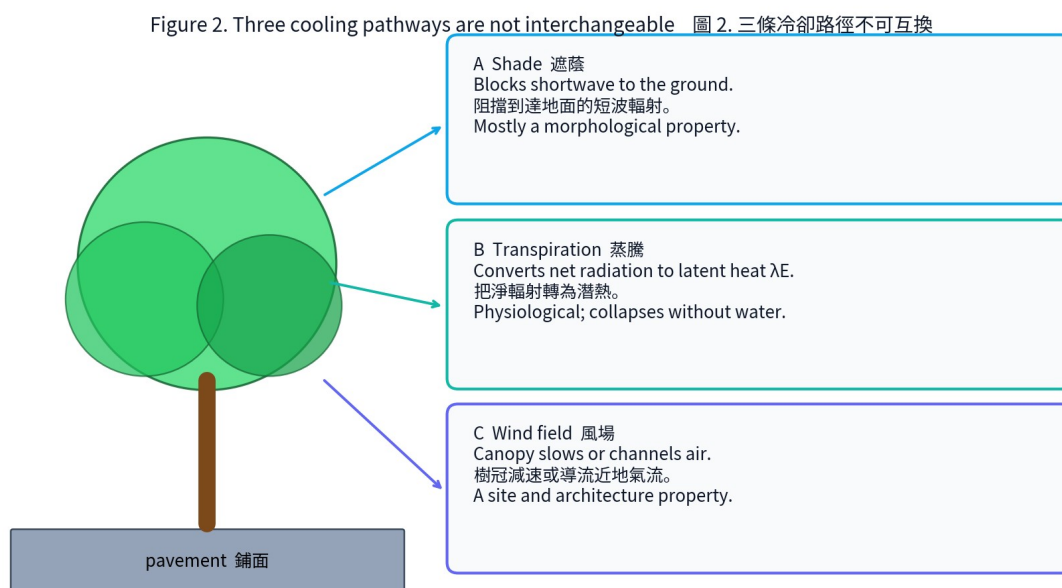


Figure 2. Shade, transpiration and wind modification are physically distinct. Only the second collapses immediately when the water account is empty. 圖 2. 遮蔭、蒸騰與風場修飾在物理上不同。只有第二條在水分賬戶清空時立即崩潰。

5. The water bill of latent-heat cooling 5. 潛熱冷卻的水賬

Latent-heat cooling is a water flux. Pataki, McCarthy, Litvak and Pincetl [5] measured sap flow on 15 species at six Los Angeles sites. August whole-tree transpiration ranged from 3.2 ± 2.3 kg tree⁻¹ d⁻¹ in unirrigated *Pinus canariensis* to 176.9 ± 75.2 kg tree⁻¹ d⁻¹ in *Platanus × hispanica* (reported as *P. hybrida*). Several widely planted taxa (*Brachychiton discolor*, *Sequoia sempervirens*, *Eucalyptus grandis*) remained below 45 kg tree⁻¹ d⁻¹. Scaled

to plots with both high per-tree rates and high planting density, transpiration reached about 2 mm d^{-1} . Winbourne et al. [12] noted that irrigated Los Angeles street trees collectively moved on the order of 30 million gallons of water per day into the atmosphere. That is not a small term in a semi-arid municipal water budget.

潛熱冷卻是水通量。Pataki、McCarthy、Litvak 與 Pincetl [5] 在洛杉磯六個地點對 15 個樹種測樹液流。八月整株蒸騰從無灌溉的 *Pinus canariensis* $3.2 \pm 2.3 \text{ kg tree}^{-1} \text{ d}^{-1}$ ，到 *Platanus × hispanica*（原文作 *P. hybrida*）的 $176.9 \pm 75.2 \text{ kg tree}^{-1} \text{ d}^{-1}$ 。若干廣植分類群（*Brachychiton discolor*、*Sequoia sempervirens*、*Eucalyptus grandis*）低於 $45 \text{ kg tree}^{-1} \text{ d}^{-1}$ 。在單株速率與種植密度皆高的樣地，蒸騰可達約 2 mm d^{-1} 。Winbourne 等 [12] 指出，洛杉磯受灌溉行道樹合計每天把約三千萬美制加侖的水送入大氣。這在半乾旱市政水預算中不是小項。

The same water flux is gated by the planting pit. Rahman et al. [4] reported mean transpiration of $4.15 \text{ g m}^{-2} \text{ min}^{-1}$ for trees over grass versus $0.44 \text{ g m}^{-2} \text{ min}^{-1}$ for trees in paved cut-outs — a factor of about ten. A specification that plants a high-transpiration species into a sealed cut-out, then claims the Los Angeles planetree number as the cooling service, is mixing two different energy balances. Lesson 17 treated irrigation as a specified trigger, dose, shut-off and term. The present review adds the reason that specification exists: without a maintained water potential, the latent-heat pathway is not delivered, and the tree may still be present as a shading object whose leaves are hotter than the air [6,7].

同一水通量被樹穴閘控。Rahman 等 [4] 報告草地上樹木平均蒸騰 $4.15 \text{ g m}^{-2} \text{ min}^{-1}$ ，鋪面開口中為 $0.44 \text{ g m}^{-2} \text{ min}^{-1}$ ——約十倍。把高蒸騰樹種種進封閉開口、再把洛杉磯懸鈴木數字當成冷卻服務的規格，是在混合兩套能量平衡。第十七課把灌溉當成須寫明觸發、劑量、關停與期限的規格。本綜述補上該規格存在的理由：沒有維持的水勢，潛熱路徑不會交付，樹仍可作為遮蔭物體存在，但其葉溫高於氣溫 [6,7]。

6. The leaf is not at air temperature 6. 葉片不處於氣溫

A leaf in the sun is an energy-balance object. Absorbed net radiation R_n is partitioned into sensible heat H , latent heat λE , storage and conduction $G + S$, and longwave exchange (Fig. 3). Stomatal closure shrinks λE ; T_{leaf} rises until H and longwave restore the balance. Leaf size, boundary-layer resistance and crown density set how tightly T_{leaf} is coupled to T_{air} [6].

陽光下的葉片是能量平衡物體。吸收的淨輻射 R_n 分配為感熱 H 、潛熱 λE 、貯存與傳導 $G + S$ ，以及長波交換（圖 3）。氣孔關閉使 λE 縮小； T_{leaf} 上升，直到 H 與長波重建平衡。葉片尺寸、邊界層阻力與樹冠密度決定 T_{leaf} 與 T_{air} 耦合有多緊 [6]。

Figure 3. Leaf energy balance and $T_{\text{leaf}} - T_{\text{air}}$ 圖 3. 葉片能量平衡與葉—氣溫差

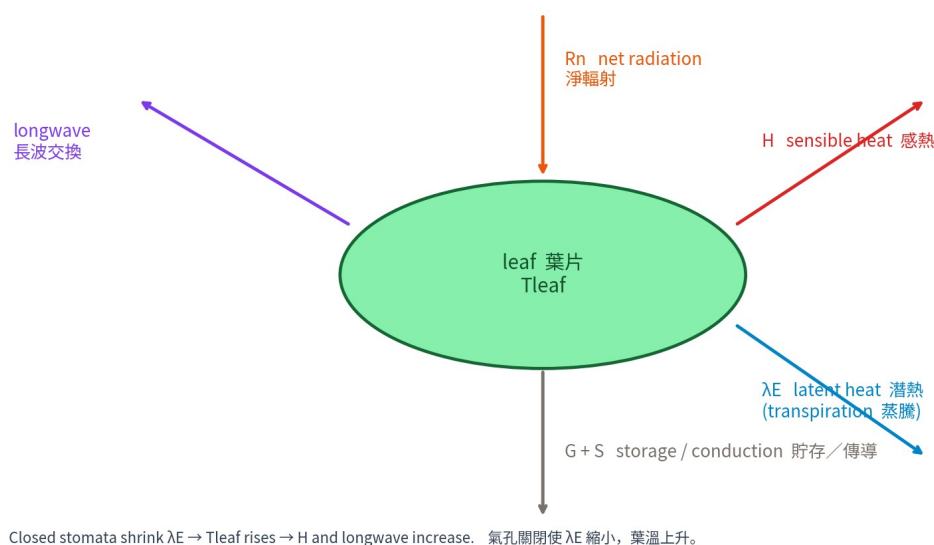


Figure 3. Schematic leaf energy balance. Closing the stomatal pathway forces the residual load into sensible heat and longwave. 圖 3. 葉片能量平衡示意。關閉氣孔路徑迫使殘餘負荷進入感熱與長波。

Leuzinger and Körner [6], working from a crane over a 35 m mixed deciduous forest at 25 °C air temperature, found that conifers and open-crowned or highly transpiring broadleaves stayed within 0.3–2.7 K of ambient, while dense-crowned Fagus, Carpinus and Tilia ran 4.5–5 K warmer, with within-crown amplitudes of 10–12 K. Calculated boundary-layer conductance was far higher for needles than for broad leaves. In Basel, Leuzinger, Vogt and Körner [7] scanned ten common urban taxa from a helicopter. Composite image peaks sat at 18 °C (water), 26 °C (vegetation), 37 °C (streets) and about 45 °C (roofs). At ~25 °C air temperature, park Aesculus crowns were near 24 °C; street Acer platanoides reached 29 °C; fifteen Norway maples exceeded 5 K above air. Across 475 trees, park trees averaged about 1 K cooler than street trees. Pinus sylvestris did not change with setting and tracked air temperature. Small-leaved taxa ran cooler than large-leaved taxa. Stomatal conductance collected under similar weather did not indicate a systematic water-supply bias between park and street, so the contrast is architectural and aerodynamic as well as physiological.

Leuzinger 與 Körner [6] 在 25 °C 氣溫下，從起重機俯瞰 35 m 混交落葉林，發現針葉樹以及開敞樹冠或高蒸騰闊葉樹維持在環境氣溫的 0.3–2.7 K 內，而密冠的山毛櫸、鵝耳櫪與椴樹則高 4.5–5 K，冠內振幅 10–12 K。計算的邊界層導度針葉遠高於闊葉。在巴塞爾，Leuzinger、Vogt 與 Körner [7] 從直升機掃描十個常見城市分類群。合成影像峰值在 18 °C（水）、26 °C（植被）、37 °C（街道）與約 45 °C（屋頂）。在約 25 °C 氣溫下，公園七葉樹樹冠近 24 °C；街道挪威槭達 29 °C；十五株挪威槭超過氣溫 5 K。475 株中，公園樹平均比行道樹涼約 1 K。歐洲赤松不隨環境改變並追隨氣溫。小葉分類群比大葉分類群涼。相似天氣下採集的氣孔導度並未顯示公園與街道之間存在系統性供水偏差，因此對比既是架構與空氣動力學的，也是生理的。

The implication for species lists is uncomfortable. A large-leaved, dense-crowned tree that is valued for shade area can run a hotter crown than a fine-leaved tree that shades less ground. Shade on pavement and cool leaves are not the same specification. Rahman et al. [4] recovered a related pattern: thin, dark leaves favoured measured cooling, but only inside a canopy-density envelope that the meta-analysis treated as primary.

這對樹種清單的含義並不舒服。因遮蔭面積被看重的大葉密冠樹，其樹冠可以比遮地較少的細葉樹更熱。鋪面上的影子與涼爽的葉子不是同一規格。Rahman 等 [4] 得到相關格局：薄而深色的葉有利測得的冷卻，但只在統合分析視為首要的樹冠密度包絡之內。

7. When the service fails 7. 服務何時失效

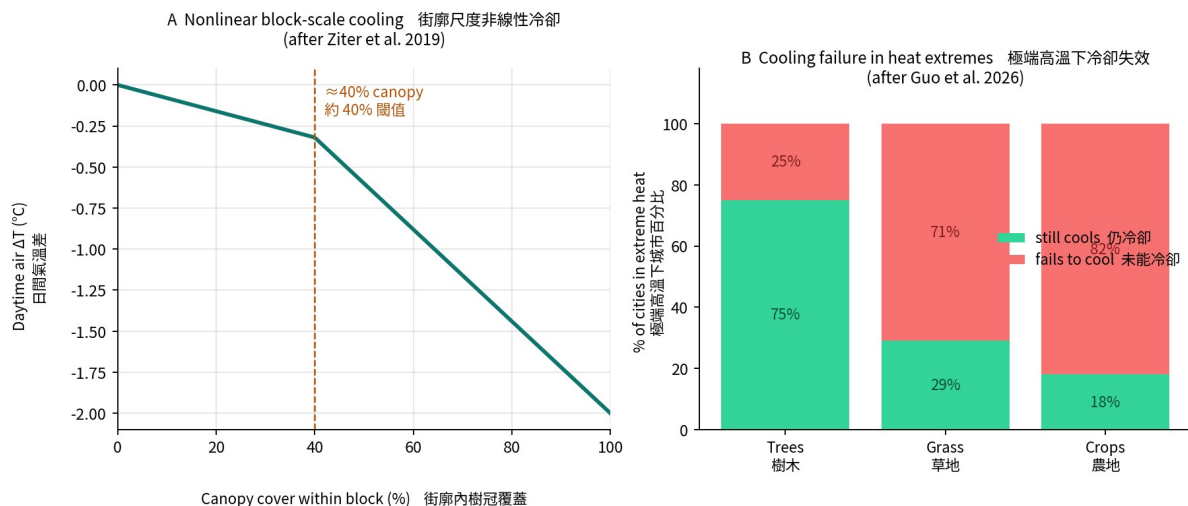


Figure 4. A: schematic of the nonlinear block-scale air-temperature response after Ziter et al. [3]. B: fraction of cities in which vegetation fails to cool during extreme heat, after Guo et al. [2]. 圖 4. A: 據 Ziter 等 [3] 的街廓尺度非線性氣溫響應示意。B: 據 Guo 等 [2], 極端高溫下植被未能冷卻的城市比例。

Failure has a defined meaning here: the vegetation surface is not cooler than the built reference, or the latent-heat pathway has collapsed, during the weather envelope when cooling is most demanded. Guo et al. [2] give the global frequencies (Fig. 4B). The physiological account is the Lesson 24 account. High VPD raises evaporative demand and, in isohydric behaviour, closes stomata to defend water potential; λE falls, T_{leaf} rises, and the tree becomes a shading body that may still lower surface temperature while adding sensible heat to the canyon air [10–12]. In

anisohydric behaviour the stomata remain more open, λE is defended longer, and the hydraulic safety margin recorded by Choat et al. [13] is spent faster. Neither strategy guarantees the service on the hottest afternoon.

此處失效有定義：在最需要冷卻的天氣包絡內，植被表面不比建成參照更涼，或潛熱路徑已崩潰。Guo 等 [2] 給出全球頻率（圖 4B）。生理賬戶即第二十四課的賬戶。高 VPD 提高蒸發需求，在等水行為中氣孔關閉以保衛水勢； λE 下降， T_{leaf} 上升，樹變成仍可能降低地表溫度、卻向峽谷空氣加入感熱的遮蔭體 [10–12]。在不等水行為中氣孔較保持開放， λE 被保衛較久，Choat 等 [13] 記錄的水力安全邊際消耗更快。兩種策略都不保證最熱下午的服務。

A second failure mode is metric error. A planting programme evaluated only on LST can report success while pedestrian air temperature and mean radiant temperature have barely moved, or the reverse [8]. A third is equity. McDonald et al. [1] found larger cooling where income and existing canopy are already higher. Expanding municipal mean canopy without concentrating cover inside hot, low-canopy blocks repeats the Ziter threshold problem at the scale of the city.

第二種失效是量測錯誤。只以 LST 評價的種植計畫可以報告成功，而行人氣溫與平均輻射溫度幾乎沒動，或相反 [8]。第三是公平。McDonald 等 [1] 發現較大冷卻出現在收入與既有樹冠本已較高之處。擴大全市平均樹冠而不把覆蓋集中到炎熱、低樹冠街廓，是在城市尺度重複 Ziter 閾值問題。

Open couplings remain. How much of the McDonald air-temperature increment is shade versus latent heat is not partitioned globally. How irrigation should be specified so that the latent-heat pathway survives the afternoon of maximum VPD is a Lesson 17 problem that this literature still treats loosely. How leaf temperature feeds back onto photosynthesis, VOC emission and pest performance is measured in fragments, not as an operational trait screen. The honest sentence for a specification is therefore conditional: a named taxon, in a named soil-volume and irrigation envelope, at a named block-scale canopy, under a named weather envelope, is expected to deliver shade continuously and latent-heat cooling only while stomata remain open.

開放耦合仍在。McDonald 氣溫增量中遮蔭與潛熱各占多少，並未在全球分割。灌溉應如何規格化，使潛熱路徑在最大 VPD 的下午存活，是第十七課的問題，本文獻仍處理得鬆。葉溫如何反饋到光合、VOC 排放與有害生物表現，只以碎片測量，尚未成為操作性狀篩檢。因此規格的誠實句子是有條件的：在具名土壤體積與灌溉包絡、具名街廓尺度樹冠、具名天氣包絡下，某一具名分類群被預期持續提供遮蔭，並只在氣孔保持開放時提供潛熱冷卻。

Practical tip 實務提示

Write cooling as two lines in the specification, not one. Line 1 is shade: crown projection at a stated solar geometry, to be delivered whenever the crown exists. Line 2 is latent heat: a water-potential or soil-moisture trigger, a dose, a shut-off, and a term, without which line 2 is not claimed. Then state the block-scale canopy that the planting is intended to help assemble. A single tree in a sealed pit cannot be asked for the city-mean number in McDonald et al. [1] or the planetree number in Pataki et al. [5].

把冷卻在規格裡寫成兩行，而不是一行。第一行是遮蔭：在給定太陽幾何下的樹冠投影，只要樹冠存在就交付。第二行是潛熱：水勢或土壤水分觸發、劑量、關停與期限；沒有這些就不主張第二行。然後寫明該種植意圖協助組成的街廓尺度樹冠。封閉樹穴中的單株，不能被要求交出 McDonald 等 [1] 的市均數字，或 Pataki 等 [5] 的懸鈴木數字。

Reflection 反思

On a street you know well, is the summer relief you feel mostly shade on the pavement, or cooler air in the canyon? Which of those two would survive a week without irrigation?

在一條你熟悉的街上，你感到的夏季緩解主要是鋪面上的影子，還是峽谷裡較涼的空氣？這兩者之中，哪一樣能在一週無灌溉之後仍然存在？

Study questions 習題

Choose the option that matches the cited result, not the popular summary.

選與被引用結果相符的選項，而不是流行摘要。

1. McDonald et al. [1] estimated that existing urban tree cover mitigates:

1. McDonald 等 [1] 估計現有城市樹冠可緩解：

- A. All mid-century climate-change warming.
- A. 本世紀中葉全部氣候暖化。
- B. 41–49% of the maximum potential air-temperature UHI that would occur with no canopy.
- B. 無樹冠時最大可能氣溫熱島的 41–49%。
- C. 40% of night-time LST in every arid city.
- C. 每一乾旱城市夜間 LST 的 40%。
- D. The need to specify irrigation.
- D. 規格化灌溉的必要。

2. The population-weighted mean summer air-temperature reduction in McDonald et al. [1] was:

2. McDonald 等 [1] 中人口加權的夏季氣溫降幅為：

- A. 3.5 °C.
- A. 3.5 °C。
- B. 0.15 ± 0.03 °C.
- B. 0.15 ± 0.03 °C。
- C. 2.7 °C in every city.
- C. 每一城市 2.7 °C。
- D. Equal to mid-century warming.
- D. 等於本世紀中葉暖化。

3. Guo et al. [2] found that during extreme heat, trees failed to cool:

3. Guo 等 [2] 發現極端高溫期間，樹木未能冷卻：

- A. 2% of cities.
- A. 2% 的城市。
- B. 25% of cities.
- B. 25% 的城市。
- C. 71% of cities.
- C. 71% 的城市。
- D. 82% of cities.
- D. 82% 的城市。

4. In Ziter et al. [3], daytime air temperature fell most steeply once canopy exceeded about:

4. 在 Ziter 等 [3] 中，日間氣溫在樹冠超過約多少後下降最陡：

- A. 10% at 10 m radius.
- A. 10 m 半徑的 10%。
- B. 40% at block scale (60–90 m).
- B. 街廓尺度（60–90 m）的 40%。
- C. 80% city-wide mean.
- C. 全市平均 80%。
- D. Zero, because the response was linear.
- D. 零，因為響應是線性的。

5. Rahman et al. [4] reported that trees over grass transpired, relative to paved cut-out pits, about:

5. Rahman 等 [4] 報告草地上的樹相對鋪面開口的蒸騰約為：

- A. The same rate.
- A. 相同速率。
- B. Half as much.
- B. 一半。
- C. Ten times as much.

- C. 十倍。
- D. One hundred times as much.
- D. 一百倍。

6. In Pataki et al. [5], August whole-tree transpiration in Los Angeles spanned roughly:

6. 在 Pataki 等 [5] 中，洛杉磯八月整株蒸騰大致跨越：

- A. 3 to 177 kg tree⁻¹ d⁻¹.
- A. 3 至 177 kg tree⁻¹ d⁻¹。
- B. 1 to 2 kg tree⁻¹ d⁻¹ for every species.
- B. 每一樹種 1 至 2 kg tree⁻¹ d⁻¹。
- C. Only conifers above 100 kg d⁻¹.
- C. 只有針葉樹高於 100 kg d⁻¹。
- D. Zero, because urban trees do not transpire.
- D. 零，因為城市樹木不蒸騰。

7. Leuzinger, Vogt and Körner [7] found that, at ~25 °C air temperature in Basel:

7. Leuzinger、Vogt 與 Körner [7] 發現，巴塞爾約 25 °C 氣溫下：

- A. All crowns equalled air temperature.
- A. 所有樹冠等於氣溫。
- B. Park Aesculus crowns were near 24 °C and some street Acer platanoides reached 29 °C.
- B. 公園七葉樹樹冠近 24 °C，部分街道挪威槭達 29 °C。
- C. Roofs were cooler than vegetation.
- C. 屋頂比植被更涼。
- D. Pinus sylvestris ran 8 K hotter in streets than in parks.
- D. 歐洲赤松在街道比公園高 8 K。

8. Shade and transpiration:

8. 遮蔭與蒸騰：

- A. Are the same flux under another name.
- A. 是同一通量的不同名稱。
- B. Can be decoupled: a droughted tree still casts shade and may add sensible heat.
- B. 可以脫鉤：受旱的樹仍投影，並可能加入感熱。
- C. Both require irrigation by definition.
- C. 兩者在定義上皆需灌溉。
- D. Are irrelevant below 40% canopy.
- D. 在樹冠 40% 以下無關。

9. Alonzo et al. [8] warned that conflicting cooling conclusions often follow from:

9. Alonzo 等 [8] 警告互相衝突的冷卻結論常常來自：

- A. Using LST versus air temperature as the metric.
- A. 以 LST 對氣溫作為量測。
- B. Measuring only conifers.
- B. 只測針葉樹。
- C. Ignoring McDonald et al. [1].
- C. 忽略 McDonald 等 [1]。
- D. Night-time bicycle sampling.
- D. 夜間自行車取樣。

10. A cooling specification that names a species but not a water-potential envelope:

10. 只點名樹種、不寫水勢包絡的冷卻規格：

- A. Still guarantees the Pataki planetree flux.
- A. 仍保證 Pataki 懸鈴木通量。

B. Claims the latent-heat pathway without the account that pays for it.

B. 在沒有支付賬戶的情況下主張潛熱路徑。

C. Is sufficient at city mean canopy of 20%.

C. 在全市平均樹冠 20% 時已足夠。

D. Cancels mid-century warming.

D. 取消本世紀中葉暖化。

Answers / 答案 1-B 2-B 3-B 4-B 5-C 6-A 7-B 8-B 9-A 10-B

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